

Panini Linguistics Olympiad 2024

Round 1 (Senior), 11th February 2024
Question Booklet

Maximum marks: 100

Total time: 4 hours

- This question booklet has 8 pages, including this page. When the exam begins, please check and inform the invigilator immediately if your booklet has less than 8 pages.
- This question booklet has 5 problems. Each problem can have multiple assignments; the marks for each assignment are indicated next to the assignment, and the total marks for the problem are indicated next to the problem title.
- **It is not expected that you will solve every problem.** It is recommended to read all the problems before starting to solve any. You don't need to solve them in order.
- At the end of some problems, we've given you some extra information and interesting facts related to the problems in footnotes. They are not required to solve the problems but may sometimes be helpful.
- All problems will ask for short objective answers (e.g. match the following, multiple choice, one-line answers, etc.). Some problems may also ask for detailed explanations of your solutions. Your answer booklet has space for both.
 - Please write all your answers in the **specific space provided** in the answer booklet.
 - Space has also been provided for answers that require explanations. If you need extra space, you may continue outside the given space, or on the back of the sheet, or on a blank sheet that the invigilator will give you. You don't need to repeat the other answers in your explanations but you can refer to them by their number if you want to.
 - You don't need to copy the questions, problem statements, or any of the data in your answers, but you can refer to them in your own way as long as it is clear.
 - Please write the **problem number**, your **roll number**, and your **name** on **every sheet** of the answer booklet, including any extra sheets.
- You are not allowed to use any calculators, mobile phones, other electronic devices, books, notes, or other external sources of information during the exam. If you have any questions at any time, you should ask the invigilator, but you are not allowed to communicate with anyone else during the exam.

All the best!

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Problem 1: Jowulu

(17 marks)

Here are some arithmetic equations in Jowulu¹. All numbers are less than 200.

$$\begin{aligned}\text{fulpon} \times \text{fulpon} &= \text{yaa bzei vapfire} \\ \text{tempun} \times \text{fuuli} &= \text{bvunfulpon} \\ \text{bvunfulpon} + \text{bzei} &= \text{kön vateë} \\ \text{bzei} \times \text{fuuli} &= \text{tontëë} \\ \text{pfire} + \text{pfire} &= \text{fulpon} \\ \text{fuuli}^{\text{teëna}} &= \text{fuuli} \\ \text{bvuntëë} \times \text{tempun} &= \text{yaa pfire vabvuntëë} \\ \text{taä}^{\text{bzei}} &= \text{yaa tontëë vataä}\end{aligned}$$

Assignment 1: (8 marks)

Rewrite the equations above with regular (Hindu-Arabic) numerals (like “1 + 1 = 2”).

Assignment 2: (6 marks)

Write in Jowulu: (a) 78; (b) 185; (c) 25.

Assignment 3: (3 marks)

Write with regular (Hindu-Arabic) numerals:

- (a) yaa tontëë vabvuntëë
- (b) yaa pfire vataä

– Animikha Dutta Dhar

Problem 2: GaeilGo Doesn't Go

(20 marks)

Everybody and everything in Ireland warmly welcomes visitors. Well, almost everything, as Aalok finds out when he decides to visit; it seems impossible to figure out the pronunciation of Gaeilge² words from their spelling! Aalok turns to his trusty pronunciation app called “GaeilGo” for help.

Given below are some Gaeilge words and their pronunciations. The pronunciations use the International Phonetic Alphabet (IPA), but knowledge of IPA is not required to solve this problem, except that the ' mark before a syllable means the syllable is stressed.

¹Jowulu, sometimes also referred to as Jɔ or Samogho, is a language of the Mande family spoken in Mali and Burkina Faso by around 10,000 to 15,000 people.

²“Gaeilge” is the Irish word for “Irish”, the native language of Ireland.

Gaeilge word	Pronunciation	Gaeilge word	Pronunciation
scuab	/s ^ʲ kuəb ^ʲ /	fluairít	/f ^ʲ l ^ʲ uə ^ʲ ˈrʲiːt̪/
scéimh	/ʃceːv̪ ^ʲ /	nialas	/ˈnʲiəl ^ʲ əs ^ʲ /
bliain	/b ^ʲ lʲiən̪ ^ʲ /	toscaire	/ˈt ^ʲ ɔs ^ʲ kə ^ʲ r̪ə/
leaid	/l̪aɪ̪ ^ʲ /	tacar	/ˈt ^ʲ akə ^ʲ r̪/
gocarsach	/ˈg ^ʲ ɔkə ^ʲ r̪s ^ʲ əx̪/	deoire	/ˈd̪iːr̪ə/
neachtar	/n̪ˈaxt ^ʲ ər̪ ^ʲ /	neodrú	/ˈn̪oːd̪r̪ ^ʲ uː/
geafáil	/ˈʲaf̪ˈaːl̪ ^ʲ /	gleo	/ʲl̪iːoː/
meacan	/ˈm̪ˈakən̪ ^ʲ /	bádchlós	/ˈb̪ˈaːd̪ˈxl̪oːs̪ ^ʲ /
tábhacht	/ˈt̪ˈaːv̪ˈəxt̪ ^ʲ /	bibe	/ˈb̪iːb̪ə/
limfe	/ˈl̪im̪ˈf̪ə/	treoluas	/ˈt̪r̪iːoːl̪uəs̪ ^ʲ /
latrach	/ˈl̪ˈat̪ˈr̪əx̪/	dobrón	/d̪ˈɔb̪ˈr̪oːn̪ ^ʲ /
sintéisach	/ʃm̪ˈt̪iːeːʃəx̪/	meáintréimhse	/m̪ˈaːn̪ˈt̪r̪iːv̪ˈj̪ə/
chéad	/ˈt̪eːd̪ ^ʲ /	feoil	/ˈf̪iːl̪ ^ʲ /
deaide	/ˈd̪ˈaːd̪ə/	frigiach	/ˈf̪r̪iːʲiːəx̪/

Assignment 1: (13 marks)

Aalok's phone is running out of battery—but perhaps he doesn't need it anymore. How should the following Gaeilge words be pronounced? In each word, the first syllable is stressed, unless an underline indicates a different stressed syllable.

Gaeilge word	Pronunciation	Gaeilge word	Pronunciation
béarla	_____ (a) _____	móráireamh	_____ (h) _____
riachtanais	_____ (b) _____	ordóige	_____ (i) _____
spiaire	_____ (c) _____	moltóirí	_____ (j) _____
débhliantúil	_____ (d) _____	deoiseach	_____ (k) _____
luachmhar	_____ (e) _____	giaráilte	_____ (l) _____
eolach	_____ (f) _____	buaibheach	_____ (m) _____
meaitseáil	_____ (g) _____		

Assignment 2: (7 marks)

Explain your solution.

– Rujul Gandhi and Shardul Chiplunkar

Problem 3: Singing it in Tlingit

(25 marks)

Given below are 21 phrases in Tlingit³ and their corresponding English translations *in arbitrary order*. Note that capitalized words in both languages are proper nouns, but not all proper nouns in Tlingit may be capitalized.⁴

A. lingit'aaní	i. around the outside of a house
B. daaká.hít	ii. someone's footprint
C. dáanaa s'aatí	iii. spring
D. dáanaa yéilaa	iv. roofing paper
E. <u>G</u> agaan Hít	v. rich person
F. gagaan wak <u>d</u> áana	vi. descendants of Raven
G. gagaan <u>x</u> 'us.eetí	vii. sunglasses
H. hít s'aatí	viii. sunlight
I. héen <u>x</u> 'aká	ix. clan leader
J. <u>k</u> aa s'aatí	x. Raven House
K. <u>k</u> aa <u>x</u> 'us.eetí	xi. Milky Way
L. Lqlaya'k' <u>x</u> 'us.eetí	xii. Sun House
M. Lqlaya'k'. <u>x</u> 'ahéeni	xiii. Winter Village
N. tàakw.àani	xiv. the world
O. tàakw.eeti	xv. envelope
P. hít kax'úx'u	xvi. Lk'ayáak'w's saliva
Q. <u>x</u> 'a	xvii. on the mouth of the water
R. x'úx' daakáx'úx'	xviii. Halloween mask
S. Yéil Hít	xix. counterfeit money
T. yéil.eetí <u>k</u> aa	xx. someone's boss
U. yéilaa	xxi. mouth

Assignment 1: (15 marks)

Determine the correct correspondences.

Assignment 2: (2 marks)

What does Lqlaya'k! x'us.eetí literally mean?

³The Tlingit language is a branch of the Na-Dene language family and is spoken by approximately 500 people in southeastern Alaska and western Canada.

⁴é, ú, í, and á are vowels; . is a glottal stop (like the sound in "uh-oh"); ! is a click consonant; x, k, and g are other consonants; and ' after a consonant indicates it is ejective, i.e. produced with a burst of air from the back of the throat.

Raven House, Sun House, and On The Water House are the names of Tlingit clans. Winter Village is the Tlingit name for Metlakatla, Annette Island. In Tlingit mythology, Raven is the creator god, described as being both a person and a raven and known as a cunning trickster. Lk'ayáak'w was a hero who chased Dry Cloud, a mythical being, for months, until they both fled up into the skies and could no longer return. A Halloween mask is a mask worn during the festival of Halloween, with the aim of concealing one's identity.

Assignment 3: (4 marks)

Translate the following Tlingit phrases literally into English. Their meanings are provided for reference.

(a) wax'ahéeni, "tears"

(b) kaa x'a.eetí, "leftover food"

Assignment 4: (4 marks)

Translate into Tlingit: (a) mayor; (b) On The Water House.

– Pratyush Vempati

Problem 4: Wiradjuri**(20 marks)**

Here are some sentences in Wiradjuri⁵ and their corresponding English translations.

inarru burai bume	a woman beat a child
bumalali nal	strike me again
nal bumani dyin	I was struck by him
bumai malang ngin-dyu	I should strike him
inarru wille dharalgiri	a woman will eat an opossum ⁶
ngin yala ngin bumali	tell him to beat him
yalai malang-nga ngin	he should have been told
bumai malang-nga nal	I should have been struck
karia ngin bumalali	let him not be struck again
karia bumalmambia	do not let him strike
bumali ngin-dyu	let me strike him
buma ngin	strike him
mirridu wille bume	a dog beat an opossum

Additionally, note that gibir means "man".

Assignment 1: (3 marks)

Translate from Wiradjuri:

(a) karia mirri dharalmambia

(b) willedu inar bumalgiri

(c) karia nal buma

⁵Wiradjuri is a language in the Pama-Nyungan family spoken by approximately 1500 people in Australia.

⁶An opossum is a nocturnal marsupial roughly the size of a house cat.

Assignment 2: (5 marks)

Translate into Wiradjuri. If a sentence has multiple possible translations, write at least one.

- (a) tell him again
- (b) eat him
- (c) I should have eaten him
- (d) a man told a child
- (e) a woman was told by an opossum

Assignment 3: (12 marks)

Explain your solution.

– Anshul Bhagwat

Problem 5: When it clicks... (18 marks)

Given below are 18 phrases in Xhosa⁷ and their corresponding English translations *in arbitrary order*.⁸

izitya ezinthathu, abantu ababini, umbuzo omnye, imibuzo emibini,
ilizwe elinye, imithi emithathu, umntu omnye, imibuzo emihlanu,
isitya esinye, ubomi, isitshixo esinye, ukutya, umfundi omnye,
izitshixo ezinthathu, ilikhaya elinye, abafundi ababini,
amakhaya amabini, ubuntu

three trees, one person, life, three dishes, one dish, two questions,
one country, one key, five questions, one student, three keys, food,
humanity, two people, one house, one question, two students,
two houses

Assignment 1: (9 marks)

Determine the correct correspondences.

Assignment 2: (3 marks)

Given that umfazi means “woman” and amazwi means “voices”, translate into Xhosa:

- (a) three women

⁷Xhosa is a Nguni language spoken by around 10 million people and is an official language in Zimbabwe and South Africa.

⁸kh, tsh, th, and ty are single consonants. x is a click consonant, i.e. produced by clicking the tongue inside the mouth. The given data have been simplified from the original Xhosa for this problem.

- (b) one voice
- (c) womanhood

Assignment 3: (6 marks)

Explain how to translate such phrases into Xhosa.

– Druhan Rajiv Shah

This is the end.